

Ordered fields, supremum/infimum and the completeness axiom



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Properties R_1 and R_2

Introduction: Properties R_1 and R_2

- ▶ We now begin to study more closely the **structure of the number sets** that we introduced earlier, and in particular: the set $\mathbb{Q}$ of **rational numbers** and the set $\mathbb{R}$ of **real numbers**.
- ▶ In this section we see the properties of the operations of **sum** (R_1) and **product** (R_2)
- ▶ From them follows the possibility of performing the four fundamental operations

Property R_1 : the sum

We denote by a, b, c three generic **rational** or **real** numbers.

$R_1 \rightarrow$ In $\mathbb{Q}$ and $\mathbb{R}$ an operation is defined (called **addition** or **sum**) which has the following four properties:

- 1 $\forall a, b, \quad a + b = b + a$ (**commutative property**)
- 2 $\forall a, b, c, \quad (a + b) + c = a + (b + c)$ (**associative property**)
- 3 there exists a **neutral element for the sum**, denoted by 0, such that:

$$\forall a, \quad a + 0 = a$$

- 4 for every a there exists an **inverse element of a with respect to the sum**, called the **opposite** of a and denoted by $-a$, such that:

$$a + (-a) = 0$$

Property R_2 : the product

$R_2 \rightarrow$ In $\mathbb{Q}$ and $\mathbb{R}$ an operation is defined (called **multiplication** or **product**) which has the following four properties:

- 1 $\forall a, b, \quad a \cdot b = b \cdot a$ (commutative property)
- 2 $\forall a, b, c, \quad (a \cdot b) \cdot c = a \cdot (b \cdot c)$ (associative property)
- 3 there exists a **neutral element for the product**, denoted by 1, such that:

$$\forall a, \quad a \cdot 1 = a$$

- 4 for every $a \neq 0$ there exists an **inverse** element of a **with respect to the product**, called the **reciprocal** of a and denoted by a^{-1} , such that:

$$a \cdot a^{-1} = 1$$

Distributive property and the four operations

- ▶ The operations of sum and product are linked by the following property:

$$\forall a, b, c, \quad (a+b) \cdot c = a \cdot c + b \cdot c \quad (\text{distributive property})$$

- ▶ From properties R_1 and R_2 follows the **possibility of performing the four fundamental operations without restrictions**:

1. **addition** and **multiplication** (defined above)
2. **subtraction**, by setting:

$$a - b = a + (-b)$$

3. **division**, by setting:

$$a/b = a \cdot b^{-1} \quad \text{provided that } b \neq 0$$

Summary

- ▶ R_1 : the sum is commutative, associative, has neutral element 0 and every a has the opposite $-a$
- ▶ R_2 : the product is commutative, associative, has neutral element 1 and every $a \neq 0$ has the reciprocal a^{-1}
- ▶ Distributive property: $(a + b) \cdot c = a \cdot c + b \cdot c$
- ▶ In $\mathbb{Q}$ and $\mathbb{R}$ sum, product, subtraction and division (for $b \neq 0$) can be performed without restrictions

Commensurable magnitudes



Introduction: Commensurable magnitudes

- ▶ In this section we introduce geometric **magnitudes** and how to compare them
- ▶ We define **homogeneous**, **commensurable** and **incommensurable** magnitudes
- ▶ Incommensurable magnitudes will show that rational numbers are not enough to measure everything

Homogeneous magnitudes

In mathematics, the word **magnitude** denotes a **property** of a **geometric figure** that can be **measured**. Examples of magnitudes are: the length of a segment, the measure of an angle, the area of a plane figure or the volume of a solid.

Definition 1: of homogeneous magnitudes

*Two magnitudes are **homogeneous** if they have the same dimension, i.e., if they can be expressed using the same unit of measurement.*

- ▶ The length of a segment and the height of a solid are two homogeneous magnitudes; indeed, both can be expressed with the same unit of length.
- ▶ The area of a square and the volume of a cone are not homogeneous magnitudes; indeed, area is expressed in square meters or in other units of surface, which however cannot be used to express volume.

Commensurable and incommensurable magnitudes

Definition 2: of commensurable and incommensurable magnitudes

*Two homogeneous magnitudes a and b are called **commensurable** if their ratio is a rational number, i.e., if $a/b \in \mathbb{Q}$. They are instead called **incommensurable** if their ratio is not a rational number, i.e., if $a/b \notin \mathbb{Q}$.*

Example

Commensurable magnitudes. The volume of a sphere and the volume of the cylinder circumscribed about the sphere are commensurable. In formulas, the volume V_s of the sphere as a function of its radius r is given by the formula:

$$V_s = \frac{4}{3} \pi r^3$$

The cylinder circumscribed about it has base radius equal to the radius r of the sphere and height h equal to twice the radius. The volume V_c of this cylinder is therefore given by the formula:

$$V_c = \pi r^2 h = \pi r^2 2r = 2\pi r^3$$

Taking the ratio we obtain:

$$\frac{V_s}{V_c} = \frac{\frac{4}{3} \pi r^3}{2 \pi r^3} = \frac{2}{3} \in \mathbb{Q}$$

Example

Incommensurable magnitudes. The length of a circle and the length of its diameter are incommensurable. In formulas, denoting by d the diameter of the circle and by c the length of the circle, we have:

$$c = \pi d$$

Now, taking the ratio between the length of the circle and its diameter, we obtain:

$$\frac{c}{d} = \frac{\pi d}{d} = \pi \notin \mathbb{Q}$$

A proof that $\pi \notin \mathbb{Q}$ will be given later.

Summary

- ▶ A **magnitude** is a measurable property of a geometric figure
- ▶ **Homogeneous** magnitudes: expressed with the same unit of measurement
- ▶ **Commensurable**: rational ratio (sphere and circumscribed cylinder: $\frac{2}{3}$)
- ▶ **Incommensurable**: non-rational ratio (circle and diameter: π)

Geometric representation of $\mathbb{Q}$

Introduction: Geometric representation of $\mathbb{Q}$

- ▶ In this section we represent rational numbers as points of a line
- ▶ Once a unit of measurement is fixed, rational numbers correspond to the points identifying segments commensurable with it

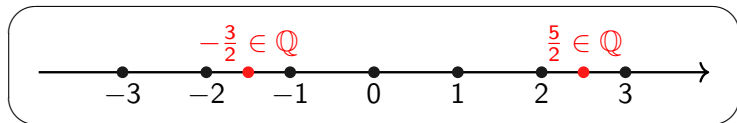
The Euclidean line

A geometric representation of $\mathbb{Q}$ can be obtained by associating with each rational number a point of the **Euclidean line** (which can be imagined as a line in the plane of infinite length)

- ▶ With an arbitrarily chosen point of the Euclidean line we associate 0, and with another point, distinct from the first, we associate 1, thus identifying the oriented segment 01, which constitutes the **unit of measurement**
- ▶ At this point we have a **one-to-one correspondence** between the rational numbers and those points P of the line that are endpoints of oriented segments $0P$ **commensurable** with 01. They are clearly commensurable since:

$$\frac{P}{1} \in \mathbb{Q} \quad \text{with} \quad P \in \mathbb{Q}$$

The Euclidean line



Total order relations

Introduction: Total order relations

- ▶ In this section we recall that “ $\leq$ ” is a total order relation on $\mathbb{Q}$ and on $\mathbb{R}$
- ▶ This allows us to compare any two numbers, the basis for defining maxima, minima and suprema

$\mathbb{Q}$ and $\mathbb{R}$ are totally ordered

Observation 1

The sets of rational numbers $\mathbb{Q}$ and of real numbers $\mathbb{R}$ with the relation “less than or equal to” (“ $\leq$ ”) are totally ordered sets.

- The relations “less than or equal to” (“ $\leq$ ”) for the set of rational or real numbers:

$$R_{\leq} = \left\{ (a, b) : a, b \in \mathbb{Q} \quad \text{and} \quad a \leq b \right\}$$

$$R_{\leq} = \left\{ (a, b) : a, b \in \mathbb{R} \quad \text{and} \quad a \leq b \right\}$$

are **partial order relations**.

Partial and total order

That is, they satisfy the following properties:

$$\forall a, \quad a \leq a \quad (\text{reflexive property})$$

$$\forall a, b, \quad a \leq b \text{ and } b \leq a \Rightarrow a = b \quad (\text{antisymmetric property})$$

$$\forall a, b, c, \quad a \leq b \text{ and } b \leq c \Rightarrow a \leq c \quad (\text{transitive property})$$

Moreover, the relations “less than or equal to” (“ $\leq$ ”) for $\mathbb{Q}$ and $\mathbb{R}$ are **total relations** since:

$$\forall a, b, \quad a \leq b \quad \text{or} \quad b \leq a$$

- Consequently, the pair consisting of the set $\mathbb{Q}$ or $\mathbb{R}$ and the corresponding relation $R_{\leq}$ is a totally ordered set.

Property R_3

Introduction: Property R_3

- ▶ In this section we highlight the properties of the ordering of rational or real numbers
- ▶ Property R_3 links the order to the operations of sum and product
- ▶ All the rules of algebraic computation follow from R_1 , R_2 , R_3

Property R_3

$R_3 \rightarrow$ In $\mathbb{Q}$ and $\mathbb{R}$ the total order relation “less than or equal to” (“ $\leq$ ”) is defined, compatible with the algebraic structure¹, that is:

1. $\forall a, b, c, \quad a \leq b \Rightarrow a + c \leq b + c$
2. $\forall a, b, c > 0, \quad a \leq b \Rightarrow a \cdot c \leq b \cdot c$

- We observe that all the rules of algebraic computation follow from the properties: R_1, R_2, R_3 .

¹an algebraic structure is a set, called the underlying set (of the structure), equipped with one or more operations

Example

For example, all the usual procedures used to solve inequalities are a consequence of the axioms and of the algebraic properties of the sum and the product, expressed by R_1 and R_2 .

Ordered fields

Introduction: Ordered fields

- ▶ In this section we define **fields** and **ordered fields**
- ▶ We will see that $\mathbb{Q}$ and $\mathbb{R}$ are both ordered fields
- ▶ But $\mathbb{Q}$ is inadequate for expressing lengths: some “empty places” remain on the line

Definition 3: of (ordered) field

An **ordered field** is a set on which two operations (sum and product) and a total order relation are defined, satisfying the properties R_1, R_2, R_3 . A set with only the properties R_1, R_2 is called a **field**.

- Everything we have said so far about the operations of sum and product and about the total order relation “less than or equal to” (“ $\leq$ ”) holds both for the set of rational numbers and for the set of real numbers.

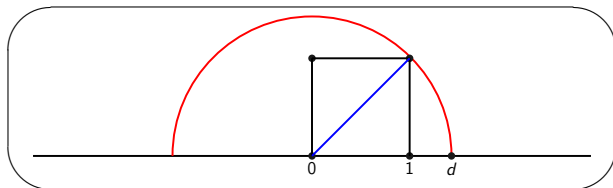
Observation 2

The sets of rational numbers $\mathbb{Q}$ and of real numbers $\mathbb{R}$ are ordered fields.

- ▶ From this point of view, therefore, $\mathbb{Q}$ and $\mathbb{R}$ appear to have similar properties. In what follows we will highlight the **property that substantially distinguishes $\mathbb{R}$ from $\mathbb{Q}$ and that makes $\mathbb{R}$ the right setting for developing mathematical analysis.**
- ▶ We begin by observing that the **set of rational numbers is inadequate for expressing the lengths of segments** (but also areas, volumes, times, speeds, etc.)

The diagonal of the square

As we have seen, there exist magnitudes that are not **commensurable** with each other. The classic example is given by the diagonal and the side of a square: if the side measures 1, the abscissa d that measures the diagonal is not a rational number.



The “empty places” on the line

Indeed, by the Pythagorean Theorem we have:

$$d^2 = 1^2 + 1^2 = 2$$

but, as we have seen, there is no rational number whose square is 2.

Hence the point d on the line does not represent any rational number. This means that, after “occupying” the points of the line with the rational numbers, there are still some **empty places** left on it.

Summary

- ▶ “ $\leq$ ” is a total order relation on $\mathbb{Q}$ and on $\mathbb{R}$
- ▶ R_3 : the order is compatible with sum and product (for $c > 0$)
- ▶ **Field:** R_1, R_2 ; **ordered field:** R_1, R_2, R_3
- ▶ $\mathbb{Q}$ and $\mathbb{R}$ are ordered fields
- ▶ The diagonal of the unit square has no rational measure: in $\mathbb{Q}$ the line has “empty places”

Bounded sets, maxima and minima

Introduction: Bounded sets, maxima and minima

- ▶ In this section we define **bounded** sets (above, below)
- ▶ We define the **maximum** and the **minimum** of a set
- ▶ With many examples we will see that a bounded set may have no maximum or no minimum

Definition 4: of bounded set

*Let E be a set contained in $\mathbb{Q}$ or in $\mathbb{R}$. The set E is called **bounded** if there exist two numbers m and M such that:*

$$\forall x \in E, \quad m \leq x \leq M$$

- ▶ E is **bounded below** if there exists a number m such that:
 $\forall x \in E, \quad m \leq x$
- ▶ E is **bounded above** if there exists a number M such that:
 $\forall x \in E, \quad x \leq M$
- ▶ It makes no difference whether we require m and/or M to belong to $\mathbb{Q}$ or to $\mathbb{R}$, since the condition only concerns the existence of such numbers.

Maximum and minimum

Definition 5: of maximum and minimum of a set

An element $x_M \in E$ is called the **maximum** of E if:

$$\forall x \in E, \quad x \leq x_M$$

An element $x_m \in E$ is called the **minimum** of E if:

$$\forall x \in E, \quad x_m \leq x$$

Maximum, minimum and boundedness (1/2)

The existence of the maximum and of the minimum of a set implies that the set is bounded, that is:

$$\text{existence of maximum and minimum} \Rightarrow \text{bounded set} \quad (1)$$

but the converse is not true (we will give a counterexample later):

$$\text{bounded set} \not\Rightarrow \text{existence of maximum and minimum} \quad (2)$$

Hence the fact that a set is bounded is a necessary but not sufficient condition for the set to have a maximum and a minimum. Moreover, the existence of a maximum and a minimum is a sufficient but not necessary condition for a set to be bounded.

Maximum, minimum and boundedness (2/2)

From the contrapositive of (1) we have:

unbounded set $\Rightarrow$ non-existence of maximum and minimum

that is, if a set is not bounded, it has no maximum or it has no minimum.

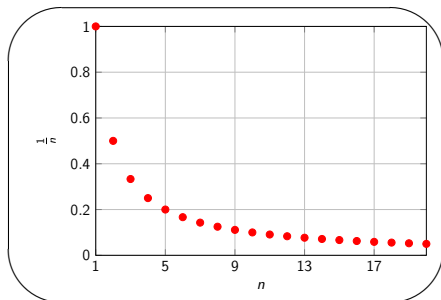
Example

Maxima and minima.

Example	set E	minimum	maximum
I)	$\mathbb{N}$	0	does not exist
II)	even integers	does not exist	does not exist

Example

Example	set E	minimum	maximum
III)	$\left\{ \frac{1}{n} : n \in \mathbb{N} \setminus \{0\} \right\}$	does not exist	1

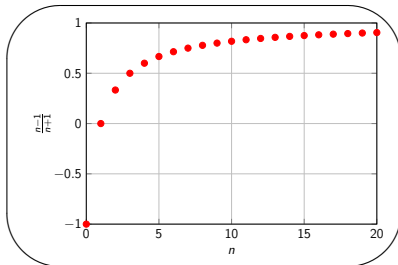


Example

Example	set E	minimum	maximum
IV)	$\left\{ \frac{n-1}{n+1} : n \in \mathbb{N} \right\}$	-1	does not exist

We have:

$$\frac{n-1}{n+1} = \frac{n+1-1-1}{n+1} = 1 - \frac{2}{n+1}$$



Bounded sets without maximum or minimum

Note that sometimes, even though the set is bounded, it may have no maximum or no minimum.

Example

Example	set E	minimum	maximum
V)	$\{x \in \mathbb{R} : 27 \leq x^3\}$	3	does not exist

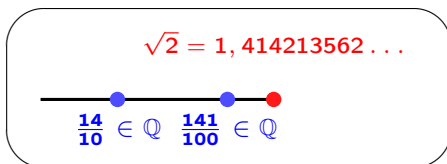
We have:

$$27 \leq x^3 \iff \sqrt[3]{27} = 3 \leq x \quad \text{and} \quad 3 \in E$$

Example

Example	set E	minimum	maximum
VI)	$\{x \in \mathbb{Q} : x \geq 0, x < \sqrt{2}\}$	0	does not exist

The value $\sqrt{2}$ is irrational, hence by truncating we obtain rational numbers that get closer and closer to $\sqrt{2}$, none of which has the property of being the maximum.

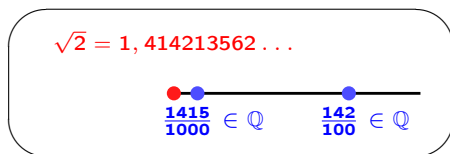


It is an example of a bounded set that has no maximum.

Example

Example	set E	minimum	maximum
VII)	$\{x \in \mathbb{Q} : x > \sqrt{2}, x \leq 4\}$	does not exist	4

The value $\sqrt{2}$ is irrational, hence by truncating and rounding up we obtain rational numbers that get closer and closer to $\sqrt{2}$, none of which has the property of being the minimum.



It is an example of a bounded set that has no minimum.

Summary

- ▶ **E bounded:** there exist m, M with $m \leq x \leq M$ for every $x \in E$
- ▶ **Maximum** (minimum): element of E greater (less) than or equal to all the others
- ▶ Maximum and minimum $\Rightarrow$ bounded, but not conversely
- ▶ Examples: $\{\frac{1}{n}\}$ has no minimum; $\{\frac{n-1}{n+1}\}$ has no maximum
- ▶ In $\mathbb{Q}$: $\{x \geq 0, x < \sqrt{2}\}$ is bounded but has no maximum

Upper/lower bounds and suprema/infima



Introduction: Upper/lower bounds and suprema/infima

- ▶ In this section we define the **upper bounds** and **lower bounds** of a set
- ▶ The **supremum** (infimum) generalizes the maximum (minimum)
- ▶ We will see that in $\mathbb{Q}$ the supremum may not exist, while in $\mathbb{R}$ it exists

Upper and lower bounds

Definition 6: of upper bound of a set

Let E be a set contained in $\mathbb{Q}$ or in $\mathbb{R}$; a number k belonging to $\mathbb{Q}$ or to $\mathbb{R}$ respectively is called an **upper bound** of E if:

$$\forall x \in E, \quad x \leq k$$

Definition 7: of lower bound of a set

Let E be a set contained in $\mathbb{Q}$ or in $\mathbb{R}$; a number k belonging to $\mathbb{Q}$ or to $\mathbb{R}$ respectively is called a **lower bound** of E if:

$$\forall x \in E, \quad k \leq x$$

Supremum and infimum

We observe that a set bounded above (below) has many upper (lower) bounds. Moreover, the upper (lower) bounds do not necessarily belong to the set itself.

Definition 8: of supremum

*The **supremum** of E (denoted by $\sup E$) is the minimum of the upper bounds of E .*

Definition 9: of infimum

*The **infimum** of E (denoted by $\inf E$) is the maximum of the lower bounds of E .*

Suprema, maxima and minima

We observe that the supremum (infimum) may not exist (unbounded sets).

We also observe that if the set has a maximum (minimum), it coincides with the supremum (infimum).

Example (1/2)

Suprema and infima.

Example	set E	$\inf E$	$\sup E$	minimum	maximum
I)	$\{x \in \mathbb{Q} : x \geq 0, x < \sqrt{2}\}$	0	does not exist	0	does not exist

We have:

$$\sqrt{2} \notin E \quad \text{and} \quad \sqrt{2} \notin \mathbb{Q}$$

hence the maximum does not exist, and neither does the sup.

Example (2/2)

Example	set E	$\inf E$	$\sup E$	minimum	maximum
II)	$\{x \in \mathbb{R} : x \geq 0, x < \sqrt{2}\}$	0	$\sqrt{2}$	0	does not exist

We have:

$$\sqrt{2} \notin E \quad \text{and} \quad \sqrt{2} \in \mathbb{R}$$

that is, the maximum does not exist, but $\sqrt{2}$ is the minimum of the upper bounds and therefore it is the sup.

Summary

- ▶ **Upper bound** (lower bound): k with $x \leq k$ ($k \leq x$) for every $x \in E$
- ▶ $\sup E$: minimum of the upper bounds; $\inf E$: maximum of the lower bounds
- ▶ If the maximum (minimum) exists, it coincides with $\sup$ ($\inf$)
- ▶ $\{x \geq 0, x < \sqrt{2}\}$: in $\mathbb{Q}$ it has no sup, in $\mathbb{R}$ it has $\sup = \sqrt{2}$

Axiomatic definition of the real numbers



Introduction: Axiomatic definition of the real numbers

- ▶ In this section we introduce the **supremum property** (R_4)
- ▶ It is the property that distinguishes $\mathbb{R}$ from $\mathbb{Q}$
- ▶ We will define $\mathbb{R}$ axiomatically and see the equivalent form with **cuts** (Dedekind axiom)

Supremum property

A totally ordered number set X is said to have the **supremum property** (least upper bound property) if:

$R_4 \rightarrow$ Every non-empty set $E \subset X$ that is bounded above has a supremum in X

- ▶ It is easy to prove that, if this property holds, then it is also true that every non-empty subset of X that is bounded below has an infimum.
- ▶ It is not required that X itself has a supremum (for example, in the case $X = \mathbb{Q}$ or $\mathbb{R}$ this is certainly false!), but that every non-empty subset of X that is bounded above has one.

$\mathbb{Q}$ and the supremum property

Example VI shows that $\mathbb{Q}$ certainly does not have the supremum property. On the other hand, $\mathbb{R}$ has this property.²

- In the **axiomatic definition** of $\mathbb{R}$, this property is part of the very definition of $\mathbb{R}$

Definition 10: (axiomatic) of the real numbers

We call $\mathbb{R}$ a set that satisfies the properties R_1, R_2, R_3, R_4 , that is, an ordered field that has the supremum property

²However, we will not present a formal proof.

Dedekind axiom

- ▶ The supremum property is also called the **Dedekind axiom**, or **axiom of continuity**, or **completeness axiom**, and it can also be stated in the following equivalent form.

Let $\{A, B\}$ be a partition of $\mathbb{R}$ (i.e., A and B are non-empty disjoint sets whose union is $\mathbb{R}$); the partition is called a **cut** (section) if:

$$\forall a \in A \text{ and } \forall b \in B \text{ we have } a < b.$$

Separating element

Then one proves that:

$R'_4 \rightarrow$ For every cut $\{A, B\}$ of $\mathbb{R}$ there exists a unique real number s (called the **separating element**) such that:

$$\forall a \in A \text{ and } \forall b \in B, \quad a \leq s \leq b$$

This separating element is $\sup A$ and $\inf B$, hence $\sup A = \inf B$

Summary

- ▶ R_4 : every non-empty set bounded above has a supremum
- ▶ $\mathbb{Q}$ does not have this property, $\mathbb{R}$ does
- ▶ $\mathbb{R}$ is an ordered field with the supremum property (R_1 – R_4)
- ▶ Equivalent form (Dedekind): every cut $\{A, B\}$ has a unique separating element $s = \sup A = \inf B$

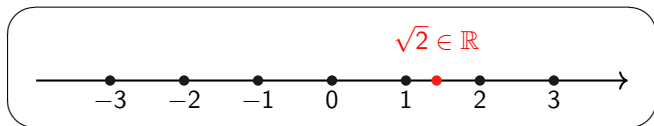
Geometric representation of $\mathbb{R}$

Introduction: Geometric representation of $\mathbb{R}$

Can the set of real numbers $\mathbb{R}$ be put in one-to-one correspondence with the points of the Euclidean line?

The real line

- ▶ Let us now return to the problem of the incommensurability of the side and the diagonal of a square. When, with straightedge and compass, having fixed the unit side of the square, we construct its diagonal and transfer it onto the starting line, the arc drawn by the compass “splits the line in two”, generating what we have called a cut; the separating element, which exists by axiom R_4 , geometrically represents the point of intersection, and therefore that number measures the diagonal.



The real line

Hence the geometric interpretation of property R_4 shows us how the set $\mathbb{R}$ is an adequate representation of our intuitive idea of a line, so adequate that **in mathematics the expression “the real line” is often used to denote $\mathbb{R}$, identifying the number set with its natural geometric representation.**

Summary (1/2)

Algebraic structure and order:

- ▶ R_1, R_2 : properties of sum and product (field)
- ▶ R_3 : total order compatible with the operations (ordered field)
- ▶ $\mathbb{Q}$ and $\mathbb{R}$ are both ordered fields

Limits of $\mathbb{Q}$:

- ▶ There exist incommensurable magnitudes (diagonal and side of a square)
- ▶ In $\mathbb{Q}$ bounded sets may have no supremum

Summary (2/2)

Suprema and infima:

- ▶ Bounded sets, maxima and minima, upper and lower bounds
- ▶ $\sup E$ = minimum of the upper bounds, $\inf E$ = maximum of the lower bounds

Real numbers:

- ▶ $\mathbb{R}$ is the ordered field with the supremum property (R_4)
- ▶ Equivalent form: Dedekind axiom (cuts and separating element)
- ▶ $\mathbb{R}$ faithfully represents the line: “the real line”